Swissquote Wallet

User journey + technical design · Four phases · 4:50 planned

Seven main slides; two Q&A backups. Read the timed script, not every line on the screen. No live recovery wait is included. Delivery time is a rehearsal target, not a measured result.

Controls: Right/Space advances, Left goes back, N shows notes, B opens Q&A. Main presentation stops at its closing slide. Never project seed words, Recovery Card words, private keys or kit codes.

0:00–0:40 · Your wallet. Your control.

We combine simple buttons with user-held keys. Create a login and wallet, add a passkey, and save the recovery material. Behind those buttons, BIP-39 generates a 24-word seed. BIP-32 derives the Ethereum signing key. Funds live at the Safe wallet address, not the key’s address. AES-256-GCM encrypts the seed on the device. Passkeys authenticate login and sensitive actions; they do not sign blockchain payments or decrypt the seed. The current prototype still needs a separate device unlock code.

0:40–1:20 · Send with confidence.

To receive, share the address on the correct network. To send, enter the recipient and amount, review, then sign. New recipients get an extra address check. The service screens the actual recipient against sanctions and issuer-freeze sources. Hits block sponsorship; unavailable required checks do not count as clear. Above five USDC in this demo, a passkey confirms the exact transaction. Local screening uses fixtures; Sepolia has live-source adapters. These controls apply to our service, not every possible blockchain submission.

1:20–2:00 · Tokens out. Fees covered.

Users should not need another token just to pay fees. On EVM, the owner signs a Safe transaction. Our funded relayer submits it and pays gas, allowing the wallet to hold zero ETH. This is direct Safe relaying, not an ERC-4337 paymaster. For Solana, we propose a separate fee payer. The user signs the transfer; the sponsor validates and co-signs the same transaction, paying SOL fees and any required token-account creation. That Solana extension is not implemented.

2:00–2:40 · Lost phone. Not lost access.

We do not split one private key between companies. The user has a payment key and an independent, twelve-word Recovery Card key. Each helper has its own key. Two recovery modules enforce the alternatives: card plus partner, or card plus Swissquote. Both signatures are required. The partner test delay is three minutes; the configured Swissquote delay is three days, not fully demonstrated. Neither helper alone, nor both together without the card, can replace the owner. Helper vault protection and identity checks are mocked.

2:40–3:25 · New key. Same wallet.

On a replacement device, generate a fresh 24-word seed and check the full new owner address. The card and helper approve replacement; the contract enforces the wait. Encrypt the replacement seed and save a fresh Emergency Kit. We use AES-256-GCM with PBKDF2-SHA-256, six hundred thousand iterations. The kit has a separate code. Finalization changes the owner, not the Safe address or balances. The old key and kit lose authority. This applies to this EVM Safe; fresh keys change ordinary Solana or Bitcoin addresses.

3:25–4:10 · Lost ≠ leaked.

A lost phone and a leaked key are different emergencies. A protected, lost key allows assisted recovery. A leaked key lets an attacker spend now, change rules, or cancel recovery. Passkeys cannot stop an external submission. If you still control the owner, replace it immediately from a clean device rather than waiting. Cancel pending recoveries in both modules and save a fresh kit. If wallet rules are compromised, move remaining assets to a new wallet if possible. Attackers can race you. Recovery cannot reverse theft.

4:10–4:50 · Built. Tested. Next: a pilot.

Our public test-network evidence shows create, pay, recover, and pay again from the same Safe address. The old key was rejected in a read-only check. Passkey tests used a virtual authenticator; recovery evidence spans retained sessions. Banking, identity and helper vault protection remain mocks; Solana remains proposed. Users get simpler payments and help after losing a device. Swissquote gets a reason to retain self-custody clients. Next: an audited, legally reviewed pilot with real users.

Q&A references

Q&A only · What changes — and what stays the same?

Technical reference only. BIP-39 generates the owner and card seeds independently. BIP-32 derives the EVM key at m/44'/60'/0'/0/0. The encryption envelope has a 16-byte random salt and 12-byte random GCM IV. Kit associated data binds chain, wallet, owner, modules and derivation path. The mnemonic-to-seed algorithm and the encryption-password derivation are separate operations. A Solana program design would need separate audited authority rotation; do not imply a plain keypair can change its key without changing its public address.

Q&A only · What the safety net does not promise.

Emergency Kit decryption works locally; settlement is not offline. A compatible Safe signing tool, current owner seed, working RPC and funded gas payer are required. The kit contains the payment seed, not a recovery share. If a Recovery Card is compromised, changing only the owner key does not replace the guardian; its rules need a separate update or migration. A malicious app can steal an unlocked seed. Passkeys are not a substitute for trustworthy software, contract audits or hardware key protection.

Evidence and scope

These slides summarize inspected source and existing evidence; they do not claim a new runtime test.

V4 acceptance: public create → pay → recover → pay-again evidence; retained sessions, not a continuous take. Old-owner GS026 refusal was read-only. Virtual WebAuthn authentication is not a physical-device test.

V4 feature walkthrough: exact button flow, separate login/device unlock, above-5-USDC passkey step-up, prepared replacement, two recovery routes, current-owner Settings rotation and Emergency Kit limits.

crypto.ts: generateOwnerSeed (256 bits), generateCardSeed (128 bits), BIP-32 derivation, AES-256-GCM and PBKDF2-HMAC-SHA-256 (600,000 iterations), encrypted kit with authenticated metadata. constants.ts: key derivation path, chain IDs and waits.

app.ts: current-owner verification, exact-hash passkey step-up, screening enforcement, sponsorship. chain.ts: direct execTransaction relay. screening.ts: public-source adapters, hit/staleness/unavailability handling.

Threat model: separate guardian keys, not MPC/Shamir shares; owner-key compromise can bypass the relay and change wallet rules. Monitor both modules. A compromised card requires guardian replacement or migration, not just owner rotation.

Solana official payment documentation: optional separate fee payer, sender signature, token accounts, transaction and account-creation costs. Solana sponsorship and stable-program-account recovery are proposals only, not features of this EVM prototype.

In the current prototype, user actions include login and device-unlock entry, separate card creation and manual backup storage. Do not present the button journey as fully automated onboarding. The owner seed can be viewed privately by locally decrypting the current kit; the wallet need not display it on the main screen. The Recovery Card is a different seed, not the payment seed.

Same address means an existing Safe deployment on the same EVM chain after owner replacement. No promise of identical deployments across EVM networks. Ordinary key-derived Solana/Bitcoin addresses change with a new key. A future program-controlled Solana design requires its own recovery policy and audit.

Gas sponsorship, sanctions screening and high-value confirmation are service policies, not a universal chain freeze. Positive screens block our relay, but an owner can use an independent gas payer. Public adapters do not establish complete legal coverage. Bank, KYC, partner identity and helper vault protection remain mocks.

Rehearsal: read the script once with a timer; aim for 4:50 maximum. Keep Q&A outside the five-minute slot. If a live demo is required, shorten narration before adding it; do not append it to this script.